Kubernetes Day 6 3 June 2026

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D (main)
$ eksctl create cluster --name prod --nodegroup-name b17dng --node-type t3.micro --nodes 5 --managed

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D (main)
$ eksctl get cluster
NAME    REGION          EKSCTL CREATED
prod    us-east-1       True

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D (main)
$ kubectl get nodes
NAME                                STATUS   ROLES    AGE     VERSION
ip-192-168-17-129.ec2.internal   Ready    <none>   4m57s   v1.34.8-eks-3385e9b
ip-192-168-21-182.ec2.internal   Ready    <none>   4m57s   v1.34.8-eks-3385e9b
ip-192-168-31-61.ec2.internal    Ready    <none>   4m57s   v1.34.8-eks-3385e9b
ip-192-168-50-167.ec2.internal   Ready    <none>   4m52s   v1.34.8-eks-3385e9b
ip-192-168-53-4.ec2.internal     Ready    <none>   4m54s   v1.34.8-eks-3385e9b

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D (main)
$ cd Kubernetes/ practice/3_June/

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ ll
total 4
-rw-r--r-- 1 Bhavana 197121 656 Jun  4 05:38 2_instadeploy_v1.yaml
-rw-r--r-- 1 Bhavana 197121   0 Jun  4 05:38 3_instadeploy_v2.yaml
```

```yaml
#2_instadeploy_v1.yaml
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: fb-deployment #name should be same in other (v2) version of deployment yaml file
  labels:
    app: fb-deploy
    env: prod

spec:
  replicas: 3
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 1
  selector:
    matchLabels:
      app: fb-app
      env: prod
  template:
    metadata:
      labels:
        app: fb-app
        env: prod
```

```
      version: v1
  spec:
    containers:
    - name: fb-container
      image: dockerhub1105/practicerepo:mc_app_img
      ports:
      - containerPort: 8080
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl apply -f 2_instadeploy_v1.yaml
deployment.apps/fb-deployment created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get all
NAME                                 READY   STATUS    RESTARTS   AGE
pod/fb-deployment-5c47c87456-gtd42   1/1     Running   0          8s
pod/fb-deployment-5c47c87456-sgbmr   1/1     Running   0          8s
pod/fb-deployment-5c47c87456-xv5gd   1/1     Running   0          8s

NAME                 TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1    <none>        443/TCP   11m

NAME                            READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/fb-deployment   3/3     3            3           9s

NAME                                       DESIRED   CURRENT   READY   AGE
replicaset.apps/fb-deployment-5c47c87456   3         3         3       10s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl describe deploy fb-deployment
Name:                   fb-deployment
Namespace:              default
CreationTimestamp:      Thu, 04 Jun 2026 05:47:52 +0530
Labels:                 app=fb-deploy
                        env=prod
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=fb-app,env=prod
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0
unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=fb-app
           env=prod
           version=v1
  Containers:
   fb-container:
    Image:         dockerhub1105/practicerepo:mc_app_img
    Port:          8080/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   fb-deployment-5c47c87456 (3/3 replicas created)
```

```
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  64s   deployment-controller  Scaled up replica
set fb-deployment-5c47c87456 from 0 to 3

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get po -o wide
NAME                                READY   STATUS    RESTARTS   AGE     IP
NODE                                NOMINATED NODE   READINESS GATES
fb-deployment-5c47c87456-gtd42   1/1     Running   0          3m59s
192.168.10.111   ip-192-168-17-129.ec2.internal   <none>           <none>
fb-deployment-5c47c87456-sgbmr   1/1     Running   0          3m59s
192.168.40.46    ip-192-168-53-4.ec2.internal     <none>           <none>
fb-deployment-5c47c87456-xv5gd   1/1     Running   0          3m59s
192.168.49.69    ip-192-168-50-167.ec2.internal   <none>           <none>

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl exec -it fb-deployment-5c47c87456-gtd42 -- bash
root@fb-deployment-5c47c87456-gtd42:/usr/local/tomcat/webapps# ll
total 32
drwxr-xr-x. 1 root root    18 Jun  4 00:18 ./
drwxr-xr-x. 1 root root    57 May  5 23:13 ../
drwxr-x---. 4 root root    54 Jun  4 00:18 ROOT/
-rw-r--r--. 1 root root 30164 May 13 09:19 ROOT.war
```

Exit

```yaml
#4_lb-svc.yaml
---
apiVersion: v1
kind: Service
metadata:
  name: fb-service
  labels:
    app: fb-service
    env: prod
spec:
  type: LoadBalancer
  selector:
    app: fb-app
    env: prod
  ports:
  - protocol: TCP
    port: 80 # This is the load balancer port, which is a external port.
    targetPort: 8080  # This is the container port on which the application is
running.
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl apply -f 4_lb-svc.yaml
service/fb-service created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/fb-deployment-5c47c87456-gtd42   1/1     Running   0          11m
pod/fb-deployment-5c47c87456-sgbmr   1/1     Running   0          11m
pod/fb-deployment-5c47c87456-xv5gd   1/1     Running   0          11m
```

```
NAME                    TYPE           CLUSTER-IP      EXTERNAL-IP
PORT(S)         AGE
service/fb-service    LoadBalancer   10.100.183.47
a288533cb3d0148ea9fe08d9dfc0b286-2042922771.us-east-1.elb.amazonaws.com
80:31672/TCP    7s
service/kubernetes    ClusterIP      10.100.0.1      <none>
443/TCP         23m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/fb-deployment       3/3     3            3           11m

NAME                                           DESIRED   CURRENT   READY   AGE
replicaset.apps/fb-deployment-5c47c87456       3         3         3       11m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get svc
NAME          TYPE           CLUSTER-IP      EXTERNAL-IP
PORT(S)         AGE
fb-service    LoadBalancer   10.100.183.47   a288533cb3d0148ea9fe08d9dfc0b286-
2042922771.us-east-1.elb.amazonaws.com   80:31672/TCP   20s
kubernetes    ClusterIP      10.100.0.1      <none>
443/TCP         23m
```


```yaml
#3_instadeploy_v2.yaml
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: fb-deployment #name should be same in other (v1) version of deployment yaml
file
  labels:
    app: fb-deploy
    env: prod
spec:
  replicas: 3
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 1
  selector:
    matchLabels:
      app: fb-app
      env: prod
  template:
    metadata:
      labels:
```

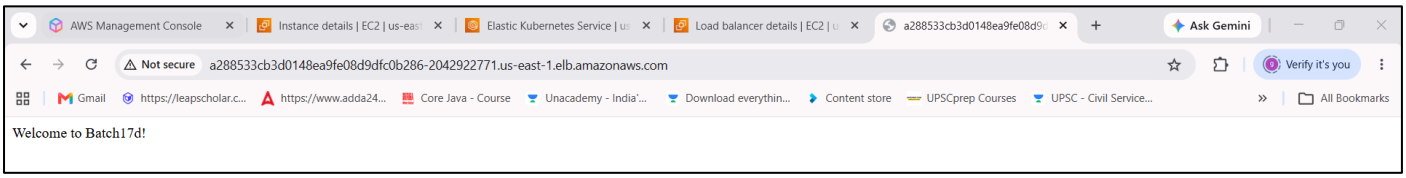
```
      app: fb-app
      env: prod
      version: v2
  spec:
    containers:
    - name: fb-container
      image: dockerhub1105/practicerepo:python_image_v1
      ports:
      - containerPort: 8080
```

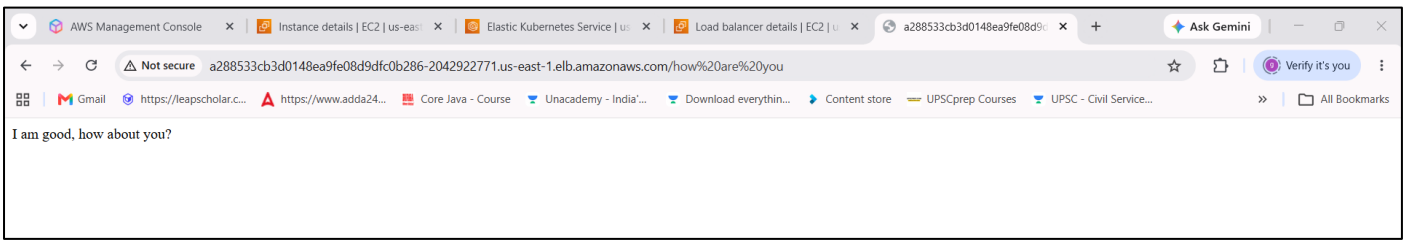
```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl apply -f 3_instadeploy_v2.yaml
deployment.apps/fb-deployment configured
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get all
NAME                                 READY   STATUS    RESTARTS   AGE
pod/fb-deployment-7f8bb96d5c-58qbz   1/1     Running   0          15s
pod/fb-deployment-7f8bb96d5c-pbd44   1/1     Running   0          15s
pod/fb-deployment-7f8bb96d5c-shvr5   1/1     Running   0          14s

NAME                  TYPE           CLUSTER-IP      EXTERNAL-IP
PORT(S)        AGE
service/fb-service    LoadBalancer   10.100.183.47
a288533cb3d0148ea9fe08d9dfc0b286-2042922771.us-east-1.elb.amazonaws.com
80:31672/TCP   6m15s
service/kubernetes    ClusterIP      10.100.0.1      <none>
443/TCP        29m

NAME                               READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/fb-deployment      3/3     3            3           17m

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/fb-deployment-5c47c87456      0         0         0       17m
replicaset.apps/fb-deployment-7f8bb96d5c      3         3         3       16s
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl describe deploy fb-deployment
Name:                   fb-deployment
Namespace:              default
CreationTimestamp:      Thu, 04 Jun 2026 05:47:52 +0530
Labels:                 app=fb-deploy
                        env=prod
Annotations:            deployment.kubernetes.io/revision: 2
Selector:               app=fb-app,env=prod
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0
unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  1 max unavailable, 1 max surge
Pod Template:
  Labels:  app=fb-app
           env=prod
           version=v2
  Containers:
   fb-container:
    Image:         dockerhub1105/practicerepo:python_image_v1
    Port:          8080/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
```

```
  Volumes:          <none>
  Node-Selectors:   <none>
  Tolerations:      <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  fb-deployment-5c47c87456 (0/0 replicas created)
NewReplicaSet:   fb-deployment-7f8bb96d5c (3/3 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  18m   deployment-controller  Scaled up replica
set fb-deployment-5c47c87456 from 0 to 3
  Normal  ScalingReplicaSet  58s   deployment-controller  Scaled up replica
set fb-deployment-7f8bb96d5c from 0 to 1
  Normal  ScalingReplicaSet  58s   deployment-controller  Scaled down replica
set fb-deployment-5c47c87456 from 3 to 2
  Normal  ScalingReplicaSet  58s   deployment-controller  Scaled up replica
set fb-deployment-7f8bb96d5c from 1 to 2
  Normal  ScalingReplicaSet  57s   deployment-controller  Scaled down replica
set fb-deployment-5c47c87456 from 2 to 1
  Normal  ScalingReplicaSet  57s   deployment-controller  Scaled up replica
set fb-deployment-7f8bb96d5c from 2 to 3
  Normal  ScalingReplicaSet  57s   deployment-controller  Scaled down replica
set fb-deployment-5c47c87456 from 1 to 0
```

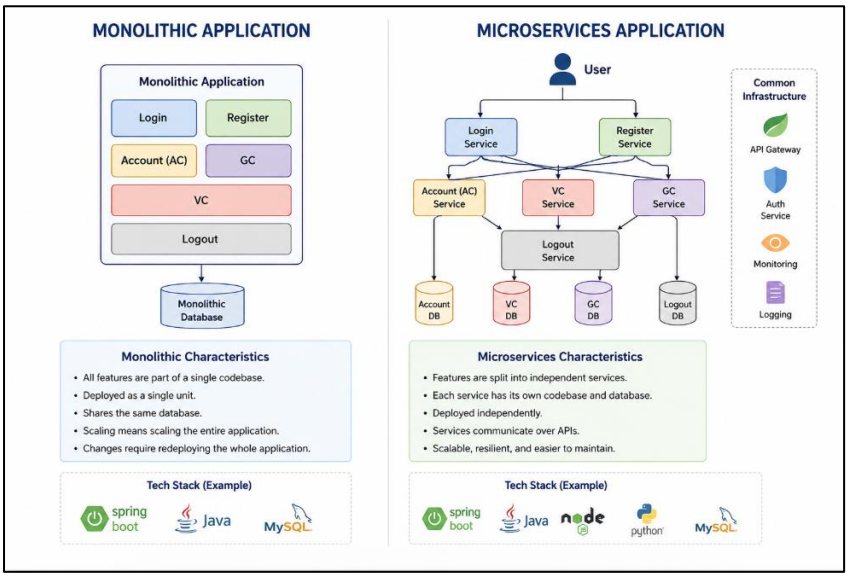


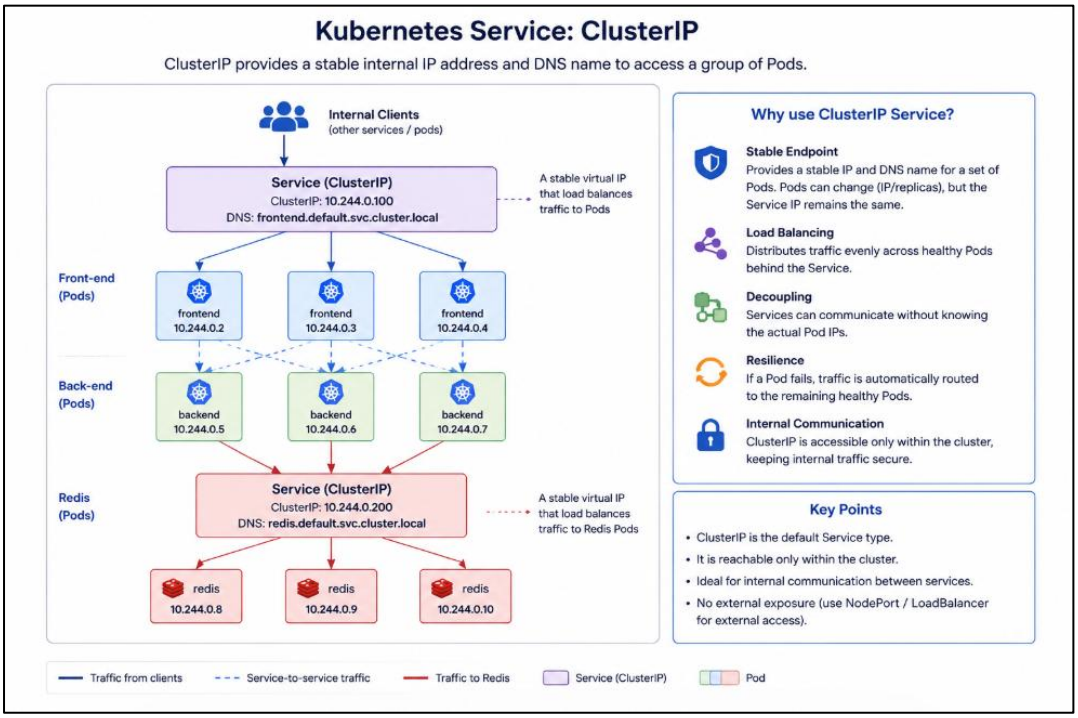



```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl rollout undo deploy fb-deployment
```


```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl delete deploy fb-deployment
deployment.apps "fb-deployment" deleted from default namespace

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get all
```

```
NAME                    TYPE           CLUSTER-IP       EXTERNAL-IP
PORT(S)        AGE
service/fb-service      LoadBalancer   10.100.183.47
a288533cb3d0148ea9fe08d9dfc0b286-2042922771.us-east-1.elb.amazonaws.com
80:31672/TCP   17m
service/kubernetes      ClusterIP      10.100.0.1       <none>
443/TCP        40m


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl delete svc fb-service
service "fb-service" deleted from default namespace
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get all
NAME                    TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes      ClusterIP   10.100.0.1   <none>        443/TCP   41m
```

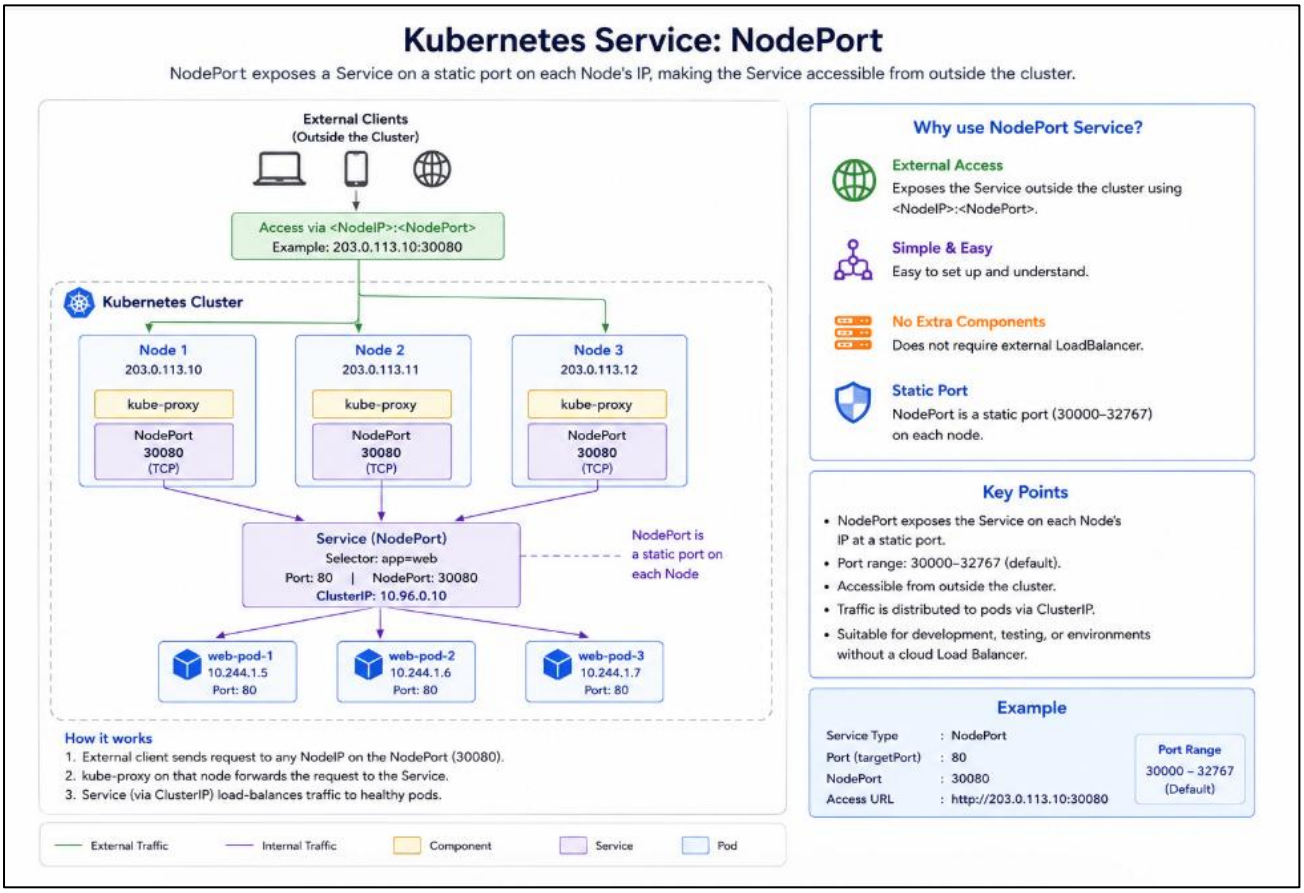




```yaml
#5_clusterip_svc.yaml
---
apiVersion: v1
kind: Service
metadata:
  name: fb-service
  labels:
    app: fb-service
    env: prod
spec:
  type: ClusterIP #we use clusterip when we want to access the service only within the cluster.
  selector:
    app: fb-app
    env: prod
  ports:
  - protocol: TCP
    port: 80 # This is the port to access the service, which is a internal port.
    targetPort: 8080  # This is the port on which the application is running inside the container.
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl apply -f 5_clusterip-svc.yaml
service/fb-service configured

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get all
NAME                 TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
service/fb-service   ClusterIP   10.100.59.42   <none>        80/TCP    27s
service/kubernetes   ClusterIP   10.100.0.1     <none>        443/TCP   60m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get svc
NAME         TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
fb-service   ClusterIP   10.100.59.42   <none>        80/TCP    50s
kubernetes   ClusterIP   10.100.0.1     <none>        443/TCP   60m


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl apply -f 3_instadeploy_v2.yaml
deployment.apps/fb-deployment created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get all
NAME                                 READY   STATUS    RESTARTS   AGE
pod/fb-deployment-58d8bd65b9-n5bzg   1/1     Running   0          9s
pod/fb-deployment-58d8bd65b9-rdrj6   1/1     Running   0          9s
pod/fb-deployment-58d8bd65b9-znrsj   1/1     Running   0          9s

NAME                 TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
service/fb-service   ClusterIP   10.100.59.42   <none>        80/TCP    114s
service/kubernetes   ClusterIP   10.100.0.1     <none>        443/TCP   61m

NAME                            READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/fb-deployment   3/3     3            3           10s
```

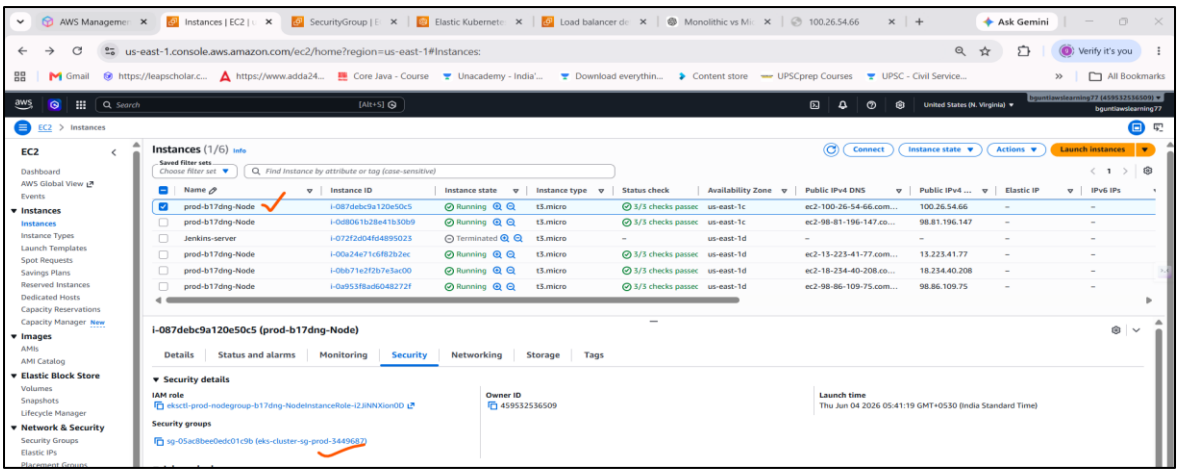
```
NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/fb-deployment-58d8bd65b9      3         3         3       10s


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl apply -f 4_lb-svc.yaml
service/fb-service configured


Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get all
NAME                                      READY   STATUS    RESTARTS   AGE
pod/fb-deployment-58d8bd65b9-n5bzg   1/1     Running   0          32s
pod/fb-deployment-58d8bd65b9-rdrj6   1/1     Running   0          32s
pod/fb-deployment-58d8bd65b9-znrsj   1/1     Running   0          32s

NAME                 TYPE           CLUSTER-IP     EXTERNAL-IP
PORT(S)        AGE
service/fb-service   LoadBalancer   10.100.59.42
ab66a2684e4df455cbdd571b4d131320-1186501111.us-east-1.elb.amazonaws.com
80:30151/TCP   2m18s
service/kubernetes   ClusterIP      10.100.0.1     <none>
443/TCP        61m

NAME                                 READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/fb-deployment   3/3     3            3           34s

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/fb-deployment-58d8bd65b9      3         3         3       34s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get svc
NAME         TYPE           CLUSTER-IP     EXTERNAL-IP
PORT(S)        AGE
fb-service   LoadBalancer   10.100.59.42   ab66a2684e4df455cbdd571b4d131320-
1186501111.us-east-1.elb.amazonaws.com   80:30151/TCP   2m31s
kubernetes   ClusterIP      10.100.0.1     <none>
443/TCP        62m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get po -o wide
NAME                                 READY   STATUS    RESTARTS   AGE   IP
NODE                                 NOMINATED NODE   READINESS GATES
fb-deployment-58d8bd65b9-n5bzg   1/1     Running   0          65s
192.168.10.111   ip-192-168-17-129.ec2.internal   <none>           <none>
fb-deployment-58d8bd65b9-rdrj6   1/1     Running   0          65s
192.168.49.69    ip-192-168-50-167.ec2.internal   <none>           <none>
fb-deployment-58d8bd65b9-znrsj   1/1     Running   0          65s
192.168.40.46    ip-192-168-53-4.ec2.internal     <none>           <none>

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl exec -it fb-deployment-58d8bd65b9-n5bzg -- bash
bash-5.2# ls
bin  boot  dev  etc  home  lib  lib64  local  media  mnt  opt  proc  root  run
sbin  srv  sys  tmp  usr  var
bash-5.2# curl 10.100.0.1:80

----- you can communicate to other service (i.e internal comm'n use cluterIp
service)
^C
bash-5.2# exit
exit
command terminated with exit code 130
```

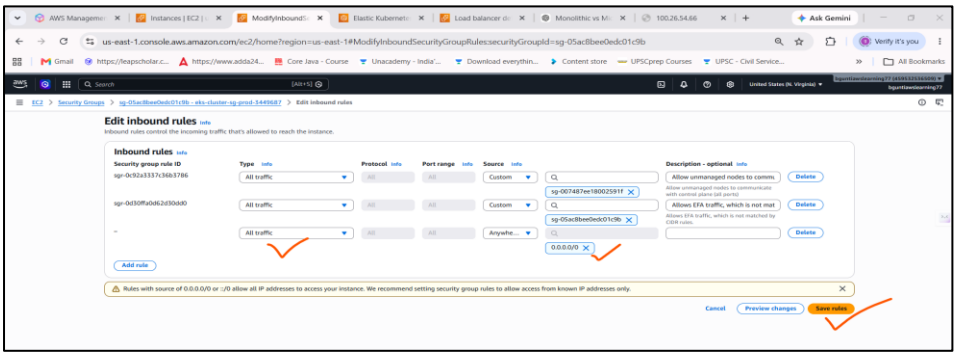


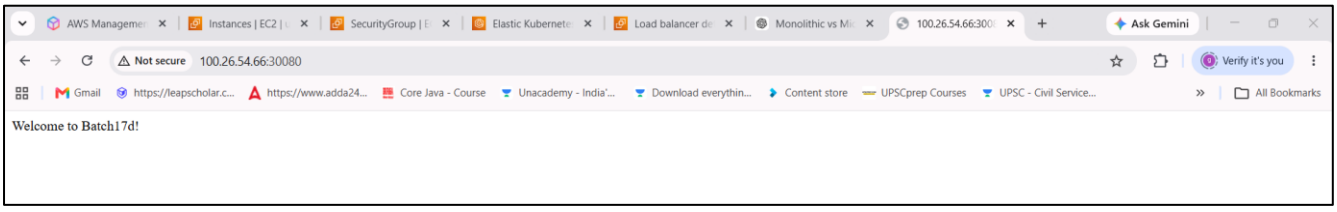

```yaml
# 6_nodeportip_svc.yaml
---
apiVersion: v1
kind: Service
metadata:
  name: fb-service
  labels:
    app: fb-service
    env: prod
spec:
  type: NodePort #we use nodeport when we want to access the service outside the cluster using node's IP and nodeport.
  selector:
    app: fb-app
    env: prod
  ports:
  - protocol: TCP
    port: 80 # This is the port to access the service, which is a external port.
    targetPort: 8080  # This is the port on which the application is running inside the container.
    nodePort: 30080 # This is the port on which we can access the service outside the cluster using node's IP and nodeport. It should be in the range of 30000-32767.
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl apply -f 6_nodeportip-svc.yaml
service/fb-service configured
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get all
NAME                                     READY   STATUS    RESTARTS   AGE
pod/fb-deployment-58d8bd65b9-n5bzg       1/1     Running   0          16m
pod/fb-deployment-58d8bd65b9-rdrj6       1/1     Running   0          16m
pod/fb-deployment-58d8bd65b9-znrsj       1/1     Running   0          16m

NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)
AGE
service/fb-service   NodePort    10.100.59.42    <none>        80:30080/TCP
18m
service/kubernetes   ClusterIP   10.100.0.1      <none>        443/TCP
77m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/fb-deployment       3/3     3            3           16m

NAME                                           DESIRED   CURRENT   READY   AGE
replicaset.apps/fb-deployment-58d8bd65b9       3         3         3       16m

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ kubectl get svc
NAME          TYPE           CLUSTER-IP      EXTERNAL-IP    PORT(S)          AGE
fb-service    NodePort       10.100.59.42    <none>         80:30080/TCP     18m
fb-service    LoadBalancer10.100.59.42    ab66a2684e4df455cbdd571b4d131320-
1186501111.us-east-1.elb.amazonaws.com   80:30151/TCP   2m31s
kubernetes    ClusterIP      10.100.0.1      <none>         443/TCP          78m
```







NameSpaces

Interview Answer (Short)
Namespaces in Kubernetes are used to logically separate resources within a cluster. They help with environment isolation (dev, test, prod), access control, resource quotas, and better organization of applications while sharing the same cluster infrastructure.

In Kubernetes, Namespaces are used to logically separate resources within the same cluster. They help organize, secure, and manage applications efficiently.
Why Do We Use Namespaces?
1. Resource Isolation
Different teams or applications can use the same cluster without interfering with each other.
Example:
Namespace: dev
  ├— nginx-pod
  ├— frontend-service

Namespace: prod
  ├— nginx-pod
  ├— frontend-service
Here, both namespaces can have resources with the same names.

---

2. Environment Separation
You can create separate environments in a single cluster.
dev
testing
staging
production
This avoids the cost of maintaining multiple clusters.

---

3. Access Control (RBAC)
Permissions can be assigned namespace-wise.
Example:
Developers → Access only dev
QA Team → Access only testing
Admins → Access all namespaces

---

4. Resource Quotas
Limit CPU, Memory, and Storage usage per namespace.
Example:
```
apiVersion: v1
kind: ResourceQuota
metadata:
  name: dev-quota
  namespace: dev
spec:
  hard:
    requests.cpu: "4"
    requests.memory: 8Gi
```
This prevents one team from consuming all cluster resources.

---

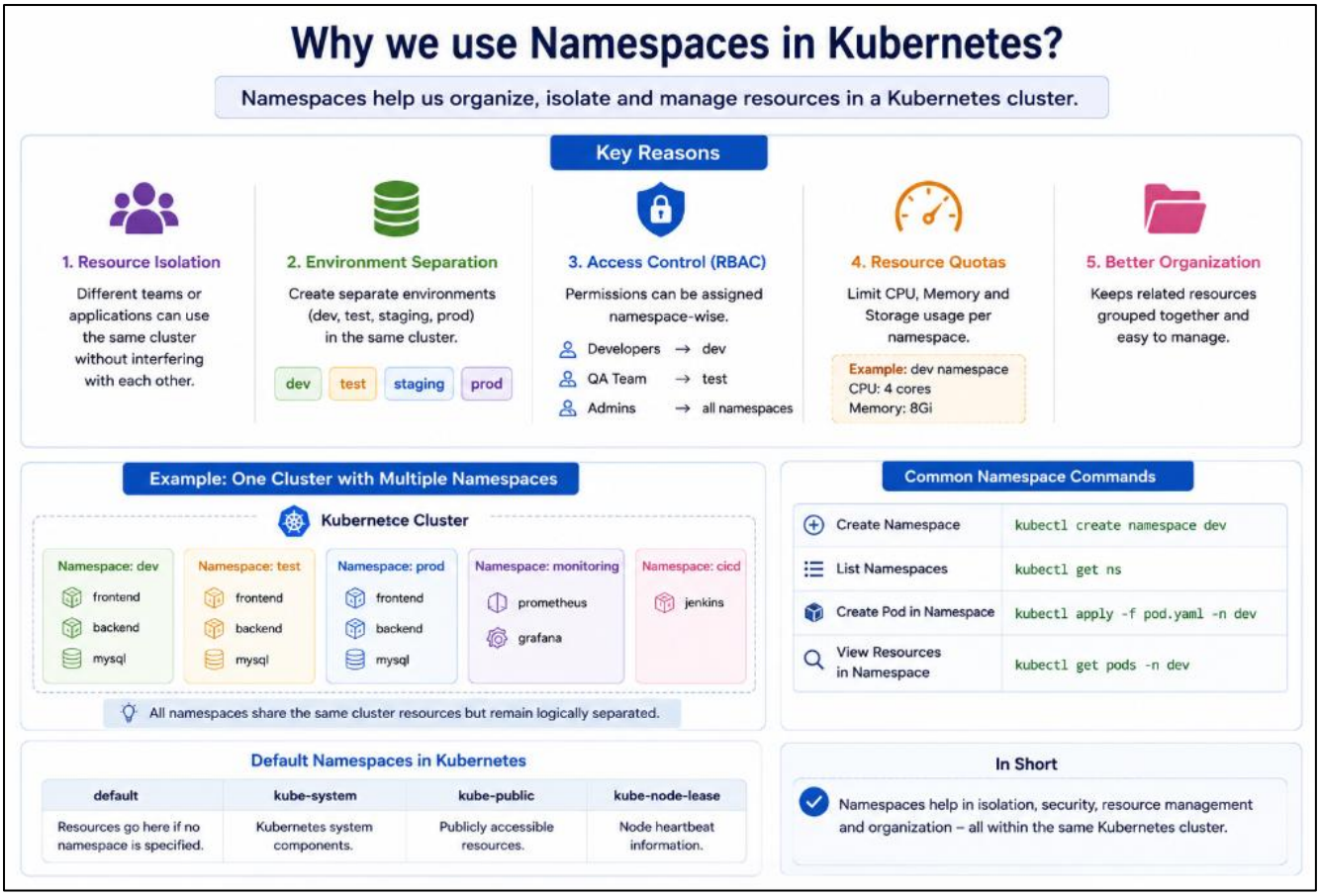
5. Better Organization
Instead of having hundreds of Pods and Services in one place:
Without Namespace:
frontend
backend

redis
mysql
jenkins
prometheus
grafana
...
With Namespace:
dev
 ├— frontend
 ├— backend

prod
 ├— frontend
 ├— backend

monitoring
 ├— prometheus
 ├— grafana

cicd
 ├— jenkins
Much easier to manage.
Real-Time Example
Suppose your company has:
Kubernetes Cluster
|
 ├— Namespace: dev
| ├— frontend
| ├— backend
| └— mysql
|
 ├— Namespace: prod
| ├— frontend
| ├— backend
| └— mysql
|
 ├— Namespace: monitoring
| ├— prometheus
| └— grafana
|
└— Namespace: cicd
    └— jenkins
All applications run in the same cluster but remain logically separated.




```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/3_June (main)
$ cd ..

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice (main)
$ cd 1_june/

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ ll
total 5
-rw-r--r-- 1 Bhavana 197121 342 Jun  2 11:46 1_pod.yaml
-rw-r--r-- 1 Bhavana 197121 455 Jun  2 11:59 2_pod.yaml
-rw-r--r-- 1 Bhavana 197121 343 Jun  2 12:34 3_multicont_pod.yaml
-rw-r--r-- 1 Bhavana 197121 612 Jun  2 13:04 4_rs.yaml
-rw-r--r-- 1 Bhavana 197121 218 Jun  2 13:45 5_pod.yaml

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl apply -f 1_pod.yaml
pod/fb-pod created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get all
NAME          READY   STATUS    RESTARTS   AGE
pod/fb-pod    1/1     Running   0          13s

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.100.0.1   <none>        443/TCP   5m8s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl apply -f 1_pod.yaml
pod/fb-pod unchanged
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl describe pod fb-pod
Name:             fb-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             ip-192-168-17-129.ec2.internal/192.168.17.129
Start Time:       Thu, 04 Jun 2026 07:27:02 +0530
Labels:           app=fb-app
                  env=prod
Annotations:      <none>
Status:           Running
IP:               192.168.10.111
IPs:
  IP:  192.168.10.111
Containers:
  fb-container:
    Container ID:
containerd://c180268d2862ebe1111dc14fb28e1f6b761d1e1640803f1a2217e1e13666388f
    Image:          tomcat
    Image ID:
docker.io/library/tomcat@sha256:f8d2287df1bb2bf9b75a6acfc6d25a75b3a9044429906f
f04e01533cfbafa028
    Port:           8080/TCP
    Host Port:      0/TCP

Events:
  Type    Reason     Age   From               Message
  ----    ------     ----  ----               -------
  Normal  Scheduled  3m7s  default-scheduler  Successfully assigned
default/fb-pod to ip-192-168-17-129.ec2.internal
  Normal  Pulling    3m7s  kubelet            spec.containers{fb-container}:
Pulling image "tomcat"
  Normal  Pulled     3m1s  kubelet            spec.containers{fb-container}:
Successfully pulled image "tomcat" in 6.01s (6.01s including waiting). Image
size: 154491880 bytes.
  Normal  Created    3m1s  kubelet            spec.containers{fb-container}:
Created container: fb-container
  Normal  Started    3m    kubelet            spec.containers{fb-container}:
Started container fb-container

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl create namespace dev
namespace/dev created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl create namespace qa
namespace/qa created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl create namespace prod
namespace/prod created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get ns
NAME              STATUS   AGE
default           Active   115m
dev               Active   34s
kube-node-lease   Active   115m
kube-public       Active   115m
kube-system       Active   115m
prod              Active   15s
qa                Active   22s
```

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl apply -f 1_pod.yaml -n qa
pod/fb-pod created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl apply -f 1_pod.yaml -n dev
pod/fb-pod created

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get po -n qa
NAME     READY   STATUS    RESTARTS   AGE
fb-pod   1/1     Running   0          37s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get po -n dev
NAME     READY   STATUS    RESTARTS   AGE
fb-pod   1/1     Running   0          26s

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D/Kubernetes/practice/1_june (main)
$ kubectl get po --all-namespaces
NAMESPACE     NAME                                READY   STATUS    RESTARTS
AGE
default       fb-pod                              1/1     Running   0
6m16s
dev           fb-pod                              1/1     Running   0
63s
kube-system   aws-node-6r6l9                      2/2     Running   0
111m
kube-system   aws-node-ggcmz                      2/2     Running   0
111m
kube-system   aws-node-kdnr6                      2/2     Running   0
111m
kube-system   aws-node-v2fjh                      2/2     Running   0
111m
kube-system   aws-node-zvl5f                      2/2     Running   0
111m
kube-system   coredns-566b9b9d-8wdkw              1/1     Running   0
114m
kube-system   coredns-566b9b9d-dphxl              1/1     Running   0
114m
kube-system   kube-proxy-6nxzw                    1/1     Running   0
111m
kube-system   kube-proxy-mmhxq                    1/1     Running   0
111m
kube-system   kube-proxy-prt69                    1/1     Running   0
111m
kube-system   kube-proxy-rqhg5                    1/1     Running   0
111m
kube-system   kube-proxy-x5bbv                    1/1     Running   0
111m
kube-system   metrics-server-694dffc97b-q89rw      1/1     Running   0
109m
kube-system   metrics-server-694dffc97b-xtqjr      1/1     Running   0
109m
qa            fb-pod
```

```yaml
---
apiVersion: v1
kind: Pod
metadata:
  name: fb-pod #pod name
  labels:
    app: fb-app # this is the label of the pod, we can use this label to select the
pod.
    namespace: prod

spec:
  containers:
  - name: fb-container #container name used in the pod
    image: tomcat
    ports:
    - containerPort: 8080 # tomcat server port.
```

kubectl apply -f pod.yaml

Verify

Check namespaces:

kubectl get ns

Check pods in prod namespace:

kubectl get pods -n dev

Expected output:

NAME READY STATUS RESTARTS AGE
nginx-pod 1/1 Running 0 1m

```
Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-Devops_MC_B17D (main)
$ cd Kubernetes/practice/

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice (main)
$ eksctl get cluster
NAME    REGION          EKSCTL CREATED
prod    us-east-1       True

Bhavana@LAPTOP-E0A1LIKM MINGW64 ~/Downloads/Bhavana Work/AWS Devops/AWS-
Devops_MC_B17D/Kubernetes/practice (main)
$ eksctl delete cluster prod
2026-06-04 07:50:36 [i]  deleting EKS cluster "prod"
2026-06-04 07:50:38 [i]  will drain 0 unmanaged nodegroup(s) in cluster "prod"
2026-06-04 07:50:38 [i]  starting parallel draining, max in-flight of 1
2026-06-04 07:50:40 [i]  deleted 0 Fargate profile(s)
2026-06-04 07:50:42 [✓]  kubeconfig has been updated
2026-06-04 07:50:43 [i]  cleaning up AWS load balancers created by Kubernetes objects of Kind
Service, Ingress, or Gateway
2026-06-04 07:50:51 [i]
2 sequential tasks: { delete nodegroup "b17dng", delete cluster control plane "prod" [async]
}
2026-06-04 07:50:51 [i]  will delete stack "eksctl-prod-nodegroup-b17dng"
2026-06-04 07:50:51 [i]  waiting for stack "eksctl-prod-nodegroup-b17dng" to get deleted
2026-06-04 07:50:52 [i]  waiting for CloudFormation stack "eksctl-prod-nodegroup-b17dng"
2026-06-04 07:51:23 [i]  waiting for CloudFormation stack "eksctl-prod-nodegroup-b17dng"
2026-06-04 07:52:10 [i]  waiting for CloudFormation stack "eksctl-prod-nodegroup-b17dng"
2026-06-04 07:53:42 [i]  waiting for CloudFormation stack "eksctl-prod-nodegroup-b17dng"
2026-06-04 07:54:19 [i]  waiting for CloudFormation stack "eksctl-prod-nodegroup-b17dng"
2026-06-04 07:55:43 [i]  waiting for CloudFormation stack "eksctl-prod-nodegroup-b17dng"
2026-06-04 07:57:49 [i]  waiting for CloudFormation stack "eksctl-prod-nodegroup-b17dng"
2026-06-04 07:58:29 [i]  waiting for CloudFormation stack "eksctl-prod-nodegroup-b17dng"
2026-06-04 07:58:30 [i]  will delete stack "eksctl-prod-cluster"
2026-06-04 07:58:31 [✓]  all cluster resources were deleted
```